Math 110 — Additional Midterm Practice

Fall 2026

An operator satisfying $T^2 = T$

Let V be a finite-dimensional real vector space, and suppose that $T : V \to V$ is linear and $T^2 = T$.

(a) (3 points) Prove that $\operatorname{range} T \cap \operatorname{null} T = \{0\}$.

(b) (3 points) Prove that $V = \operatorname{range} T + \operatorname{null} T$.

(c) (4 points) Let $S : V \to V$ be a linear operator such that $ST = TS$. Prove that both $\operatorname{range} T$ and $\operatorname{null} T$ are invariant under S. Deduce that there is a basis of V with respect to which the matrix of S is block diagonal, with one block corresponding to $\operatorname{range} T$ and the other to $\operatorname{null} T$.